MODULE 4 ACCESSORY DIGESTIVE ORGANS

The Accessory Digestive Organs.

ACCESSORY ORGANS OF DIGESTION

Covers the teeth and tongue, the salivary glands, the liver, the gallbladder and the path of bile, and the pancreas. The focus is on the structures and the job each one does.

BY THE END

1. Name the six accessory digestive organs and the role of each.
2. Identify the structures of the liver, gallbladder, and pancreas.
3. Trace the path of bile from the liver to the small intestine.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Accessory digestive organs

drsrennie-stack.github.io/new-build-bio4-solano/accessory-digestive-concept-videos.html

The Accessory Organs, an Overview

The accessory digestive organs are not part of the food tube itself. The teeth and tongue work on food in the mouth; the rest add secretions to the canal through ducts.

THE SIX ACCESSORY ORGANS AND HOW THEY WORK

Structure	What it is
Accessory digestive organs	The teeth, tongue, salivary glands, liver, gallbladder, and pancreas
Teeth and tongue	The accessory organs that make direct contact with food in the mouth
Glandular accessory organs	The salivary glands, liver, and pancreas, which produce secretions delivered through ducts
Storage organ	The gallbladder, which stores and concentrates bile until it is needed

The Teeth and Tongue

The teeth and the tongue prepare food in the mouth, breaking it down and moving it for chewing and swallowing.

THE STRUCTURES OF THE TEETH AND THE TONGUE

Structure	What it is
Teeth	The accessory organs that cut and grind food during chewing
Tongue	A digestive organ of skeletal muscle covered with mucous membrane; it forms the floor of the mouth
Extrinsic tongue muscles	Muscles that originate outside the tongue and move it side to side and in and out
Intrinsic tongue muscles	Muscles within the tongue that alter its shape and size for speech and swallowing
Lingual frenulum	The midline fold of mucous membrane anchoring the tongue to the floor of the mouth
Papillae	The projections on the tongue surface that carry taste buds and aid the grip on food
Lingual glands	Small glands in the tongue that secrete mucus and watery fluid

The Salivary Glands

Saliva keeps the mouth moist, cleanses it, and begins the breakdown of food. Three large pairs of glands produce most of it. Compare them.

THE THREE LARGE PAIRS OF SALIVARY GLANDS

Gland	Location
Parotid glands	Anterior and inferior to the ears, between the skin and the masseter muscle; the duct opens by the second upper molar
Submandibular glands	In the floor of the mouth, along the inner side of the mandible; the ducts open near the lingual frenulum
Sublingual glands	Beneath the tongue, above the submandibular glands; the ducts open onto the floor of the mouth

THE MINOR GLANDS AND THEIR PRODUCT

Structure	What it is
Minor salivary glands	Small labial, buccal, palatal, and lingual glands scattered through the lining of the mouth
Saliva	The watery secretion that moistens, cleanses, and lubricates food and begins its breakdown

The Liver

The liver is the largest gland in the body. It sits below the diaphragm, mostly in the upper right abdomen, and it makes bile.

THE LOBES, LIGAMENTS, AND MICROSCOPIC STRUCTURES OF THE LIVER

Structure	What it is
Liver	The large gland below the diaphragm; it produces bile
Right and left lobes	The two principal lobes of the liver, separated by the falciform ligament
Quadrate and caudate lobes	Two smaller lobes on the posterior and inferior surface
Falciform ligament	The fold of peritoneum that suspends the liver from the anterior abdominal wall and diaphragm
Ligamentum teres	The round ligament; a fibrous cord in the free edge of the falciform ligament, a remnant of the fetal umbilical vein
Hepatocytes	The functional cells of the liver, arranged in plates called hepatic laminae
Hepatic sinusoids	The wide, leaky capillaries that run between the plates of hepatocytes
Bile canaliculi	The tiny channels between hepatocytes that collect the bile they make
Portal triad	The trio found at the corner of each liver lobule: a bile duct, a branch of the hepatic artery, and a branch of the hepatic portal vein

KEY POINT

The portal triad is three things at the corner of every lobule, and the traffic runs both ways: the branch of the hepatic artery and the branch of the hepatic portal vein bring blood in, while the bile duct carries bile out.

The Gallbladder and the Path of Bile

The gallbladder is a small sac on the underside of the liver that stores bile. Bile travels a branching set of ducts to reach the small intestine.

THE GALLBLADDER AND THE BILE DUCTS

Structure	What it is
Gallbladder	The pear-shaped sac on the posterior surface of the liver that stores and concentrates bile
Fundus, body, and neck	The three regions of the gallbladder, from the broad inferior end to the tapered superior end
Cystic duct	The duct that carries bile to and from the gallbladder
Common bile duct	The duct formed where the common hepatic duct and the cystic duct join
Hepatopancreatic ampulla	The ampulla of Vater, the short widened duct where the bile duct and pancreatic duct meet before the duodenum

Bile drains away from the liver cells along a path of widening ducts to reach the duodenum.

1. **Bile canaliculi** tiny channels between hepatocytes collect the bile

2. **Hepatic ducts** the bile is gathered into the right and left hepatic ducts, which merge into the common hepatic duct
3. **Cystic duct and gallbladder** bile may detour into the gallbladder for storage through the cystic duct
4. **Common bile duct** the common hepatic and cystic ducts join to form the common bile duct
5. **Hepatopancreatic ampulla** the common bile duct joins the pancreatic duct at the ampulla
6. **Duodenum** bile enters the small intestine at the major duodenal papilla, past the sphincter of Oddi

The Pancreas

The pancreas lies behind the stomach. It is both an exocrine gland, secreting into the duodenum, and an endocrine gland.

THE REGIONS, DUCTS, AND CLUSTERS OF THE PANCREAS

Structure	What it is
Pancreas	The gland posterior to the stomach, retroperitoneal in position
Head, body, and tail	The three regions of the pancreas, from the wide end in the curve of the duodenum to the tapered end
Pancreatic duct	The main duct, also called the duct of Wirsung; it joins the common bile duct
Accessory duct	The smaller duct, also called the duct of Santorini, that empties separately into the duodenum
Major duodenal papilla	The raised opening in the duodenal lining where the hepatopancreatic ampulla delivers bile and pancreatic juice
Sphincter of Oddi	The muscular ring that controls flow through the hepatopancreatic ampulla into the duodenum
Acini	The exocrine clusters of the pancreas that produce pancreatic juice; about 99 percent of the gland
Pancreatic islets	The endocrine clusters of the pancreas that produce hormones; about 1 percent of the gland

KEY POINT

Each pancreatic duct carries two names. The main one is the duct of Wirsung and the smaller one is the duct of Santorini, and only the main one joins the common bile duct.

Common Disorders of the Accessory Organs

Compare the common disorders of the accessory organs by the structure each one affects.

THE COMMON DISORDERS OF THE ACCESSORY ORGANS

Disorder	What it is
Gallstones	Hardened deposits that form in the gallbladder and can block the bile ducts
Cholecystitis	Inflammation of the gallbladder, often caused by a blocking gallstone
Hepatitis	Inflammation of the liver, commonly from a viral infection
Cirrhosis	The replacement of healthy liver tissue with scar tissue, with loss of liver function
Jaundice	A yellowing of the skin and eyes from a buildup of bilirubin, often from liver or bile duct disease
Pancreatitis	Inflammation of the pancreas
Mumps	A viral infection that inflames and enlarges the parotid glands

SHARED DUCTS, SHARED PROBLEMS

The bile duct and the pancreatic duct meet at the **hepatopancreatic ampulla** before they reach the duodenum. Because they share that final passage, a gallstone lodged at the ampulla can block bile and pancreatic juice at once, which is one way gallstones trigger pancreatitis.

The liver's blood supply is unusual. It receives nutrient-rich blood from the **hepatic portal vein** as well as oxygenated blood from the **hepatic artery**. That heavy blood flow filters the body's nutrients, but it also makes the liver a common site for cancer to spread to.